

CIRCLE

- The locus of a point moving in a plane such that it is at a constant distance from a fixed point is called as circle. The fixed point is known as centre and fixed distance is called as radius of the circle.
- **Important concepts on circle:**
 - (a) Angle in a semicircle is a right angle.
 - (b) The angle made by a chord of a circle at the centre is double the angle made by the chord at any point on the circumference of the circle lying on the same side of it.
 - (c) Through three non collinear points there exist one and only one circle.
 - (d) The line joining mid point of a chord and centre of the circle is perpendicular to the chord.
 - (e) Perpendicular bisector of a chord of a circle passes through the centre of the circle.
 - (f) Two tangents can be drawn from an external point to a circle.
 - (g) Only one tangent can be drawn at a given point of the circle.
 - (h) No real tangent can be drawn from an external point to a circle.
 - (i) A line is a tangent to a circle if the length of the perpendicular from centre to that line is equal to radius.
 - (j) If a quadrilateral is inscribed in a circle, the opposite angles are supplementary.
 - (k) The chord of a circle with maximum length is diameter of the circle.
 - (l) Circles with same centre are called as concentric circles.
- Equation of circle with centre (h, k) and radius r is $(x - h)^2 + (y - k)^2 = r^2$.
- Equation of circle with centre $(0, 0)$ and radius r is $x^2 + y^2 = r^2$.
- Equation of every circle is:
 - (a) Second degree in x and y and in it.
 - (b) Coefficient of $x^2 =$ Coefficient of y^2 .
 - (c) Coefficient of $xy = 0$.
- Standard equation of circle $x^2 + y^2 + 2gx + 2fy + c = 0$.
 - (a) If $g^2 + f^2 - c > 0$, then it is a real circle.
 - (b) If $g^2 + f^2 - c = 0$, then it is a point circle.
 - (c) If $g^2 + f^2 - c < 0$, then it is an imaginary circle.
- If $x^2 + y^2 + 2gx + 2fy + c = 0$ represent a circle then:
 - (a) Its radius = $\sqrt{g^2 + f^2 - c}$.
 - (b) Its centre is $(-g, -f)$.
- The equation of circle $x^2 + y^2 + 2gx + 2fy + c = 0$ contains three arbitrary constants, hence to fix a circle at least three condition are required.
- The general equation of a circle $ax^2 + ay^2 + 2gx + 2fy + c = 0$, its centre is $\left(\frac{-g}{a}, \frac{-f}{a}\right)$ and
 radius = $\frac{\sqrt{g^2 + f^2 - ac}}{|a|}$.

- The equation of a circle passing through origin and making intercepts a and b on x and y axes respectively is $x^2 + y^2 - ax - by = 0$.
- The equation of circle passing through origin and through the point of intersection of the line $ax + by + c = 0$ with co-ordinate axes is $ab(x^2 + y^2) + c(bx + ay) = 0$.
- The equation of circle with (x_1, y_1) and (x_2, y_2) are extremities of a diameter is $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$ or $x^2 + y^2 - x(x_1 + x_2) - y(y_1 + y_2) + x_1x_2 + y_1y_2 = 0$.
- Equation of circle touching:
 - (a) x -axis is of the form $x^2 + y^2 + 2gx + 2fy + g^2 = 0$, with radius $|f|$.
 - (b) y -axis is of the form $x^2 + y^2 + 2gx + 2fy + f^2 = 0$, with radius $|g|$.
 - (c) Both the axes is of the form $x^2 + y^2 + 2gx + 2fy + c = 0$, $c = g^2 = f^2$, with radius $|f| = |g|$.
- Equation of circle with centre (a, b) and touching:
 - (a) x -axis is of the form $x^2 + y^2 - 2ax - 2by + a^2 = 0$, with radius $|b|$.
 - (b) y -axis is of the form $x^2 + y^2 - 2ax - 2by + b^2 = 0$, with radius $|a|$.
 - (c) Both the axes with radius r is of the form $x^2 + y^2 \pm 2rx \pm 2ry + r^2 = 0$.
- If 'C' is the centre 'r' is radius of a circle and the perpendicular distance from centre C to a line 'l' is 'd', then:
 - (a) If $d > r \Leftrightarrow l$ is outside of the circle.
 - (b) If $d = r \Leftrightarrow l$ is a tangent (touches the circle).
 - (c) If $d < r \Leftrightarrow l$ is inside of the circle.
- **Condition for a line touches the circle:**
 - (a) $y = mx + c$ to touch circle $x^2 + y^2 = r^2$ is $c^2 = r^2(m^2 + 1)$.
 - (b) $lx + my + n = 0$ to touch circle $x^2 + y^2 = r^2$ is $n^2 = r^2(m^2 + l^2)$ and the point of contact is $\left(\frac{-lr^2}{n}, \frac{-mr^2}{n} \right)$.
 - (c) $lx + my + n = 0$ to touch circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $(l^2 + m^2)(g^2 + f^2 - c) = (lg + mf - n)^2$.
- For the circle $x^2 + y^2 = r^2$ tangent with slope 'm' is $y = mx \pm r\sqrt{m^2 + 1}$.
- For the circle $(x - x_1)^2 + (y - y_1)^2 = r^2$
 - (a) Tangent parallel to x -axis is: $y = y_1 \pm r$.
 - (b) Tangent parallel to y -axis is: $x = x_1 \pm r$.
 - (c) Tangent parallel to $lx + my + n = 0$ is: $lx + my = lx_1 + my_1 \pm r\sqrt{l^2 + m^2}$.
- The intercept made by the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ on:
 - (a) x -axis is: $2\sqrt{g^2 - c}$.
 - (b) y -axis is: $2\sqrt{f^2 - c}$.
 - (c) Intercepts are always positive.

- (d) If the circle touches x-axis then, intercept on x-axis is zero, then $c = g^2$.
- (e) If the circle touches y-axis then, intercept on y-axis is zero, then $c = f^2$.
- (f) If the circle touches both the axes then $c = g^2 = f^2$.
- If 'r' is radius of a circle then a line which is at a distance 'd' from centre of the circle cuts (a chord) an intercept of length $2\sqrt{r^2 - d^2}$.
- The number of circles touching all the three given lines, which are:
 - (a) Forming a triangle is FOUR.
 - (b) Such that two of the lines are parallel is TWO.
 - (c) Concurrent is ONE.
 - (d) All parallel is ZERO.
- In a circle, normal at any point of the circle passes through its centre.
- **Parametric form of the circle:**
 - (a) $x^2 + y^2 = r^2$ is $x = r \cos^{-}$, $y = r \sin^{-}$ and point on circle is $(r \cos^{-}, r \sin^{-})$.
 - (b) $(x - x_1)^2 + (y - y_1)^2 = r^2$ is $x = x_1 + r \cos^{-}$, $y = y_1 + r \sin^{-}$ and point on the circle is $(x_1 + r \cos^{-}, y_1 + r \sin^{-})$.
- The equation of tangent at $(r \cos^{-}, r \sin^{-})$ to the circle $x^2 + y^2 = r^2$ is $x \cos^{-} + y \sin^{-} = r$.
- The equation of normal at $(r \cos^{-}, r \sin^{-})$ to the circle $x^2 + y^2 = r^2$ is $x \cos^{-} - y \sin^{-} = 0$.
- The equation of normal at (x_1, y_1) to the circle $x^2 + y^2 = r^2$ is $xy_1 - yx_1 = 0$.
- The equation of normal at (x_1, y_1) to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $(x + g)y_1 - (y + f)x_1 = 0$.
- Equation of the chord joining the two points $(r \cos^{-}_1, r \sin^{-}_1)$ and $(r \cos^{-}_2, r \sin^{-}_2)$ of the circle $x^2 + y^2 = r^2$ is $x \cos\left(\frac{-_1 + -_2}{2}\right) + y \sin\left(\frac{-_1 + -_2}{2}\right) = r \cos\left(\frac{-_1 - -_2}{2}\right)$.
- The length of chord $\overline{AB}$ joining $A(r \cos^{-}_1, r \sin^{-}_1)$ and $B(r \cos^{-}_2, r \sin^{-}_2)$ of the circle $x^2 + y^2 = r^2$ is $2r|\sin(\frac{-_1 - -_2}{2})|$.
- If $x^2 + y^2 + 2gx + 2fy + c = 0$ be a circle and (x_1, y_1) be a point, then:
 - (a) If $x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c = 0 \Rightarrow P(x_1, y_1)$ lies on the circle.
 - (b) If $x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c > 0 \Rightarrow P(x_1, y_1)$ lies outside the circle.
 - (c) If $x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c < 0 \Rightarrow P(x_1, y_1)$ lies inside the circle.
- **Equation of circle passes through three points:** Circle passes through three points (x_1, y_1) , (x_2, y_2) , (x_3, y_3) is

$$\begin{vmatrix} x^2 + y^2 & x & y & 1 \\ x_1^2 + y_1^2 & x_1 & y_1 & 1 \\ x_2^2 + y_2^2 & x_2 & y_2 & 1 \\ x_3^2 + y_3^2 & x_3 & y_3 & 1 \end{vmatrix} = 0.$$

- **Power of a point:** If 'C' is the centre and 'r' is the radius of a circle then power of a point P is defined as $CP^2 - r^2$. **OR** If a secant through P cuts a circle in any two points A and B, then $PA \times PB$ is called as power of P.
- If $x^2 + y^2 + 2gx + 2fy + c = 0$ is a circle in standard form and a point $P(x_1, y_1)$, then the power of P is $x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$.
 - (a) Power of a point on the circle is ZERO.
 - (b) If Power is POSITIVE then point lies outside the circle.
 - (c) If Power is NEGATIVE then point lies inside the circle.
- If a point $P(x_1, y_1)$ lying outside the circle $x^2 + y^2 + 2gx + 2fy + c = 0$, then the length of the tangent from P is $\sqrt{x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c}$.
- The equation of chord joining the two points (x_1, y_1) and (x_2, y_2) of a circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is

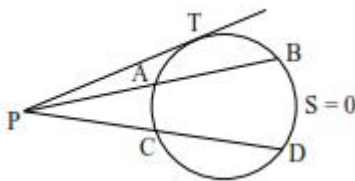
$$\Rightarrow xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = xx_2 + yy_2 + g(x + x_2) + f(y + y_2) + c$$

$$= x_1x_2 + y_1y_2 + g(x_1 + x_2) + f(y_1 + y_2) + c$$
- The equation of tangent at (x_1, y_1) to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is :

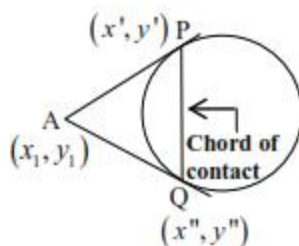
$$\Rightarrow x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c = 0$$
- **Pair of tangent from a point to the circle:** Equation of pair of tangent from a point (x_1, y_1) to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is:

$$(x^2 + y^2 + 2gx + 2fy + c)(x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c) = (x_1x + y_1y + g(x_1 + x) + f(y_1 + y) + c)^2$$
- **Director Circle:** The locus of a point of intersection of two perpendicular tangents to a circle is called as the director circle.
 - (1) Director circle of the circle $x^2 + y^2 = a^2$ is $x^2 + y^2 = 2a^2$.
 - (2) Director circle of the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is

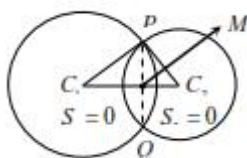
$$x^2 + y^2 + 2gx + 2fy + 2c - g^2 - f^2 = 0$$
- **Power of a point with respect to a circle:** Let $P(x_1, y_1)$ be a point on the secant PAB, drawn. The power of V with respect to the circle $S = x^2 + y^2 + 2gx + 2fy + c = 0$ is equal to the $PA \cdot PB$ i.e. $\Rightarrow x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c = 0$



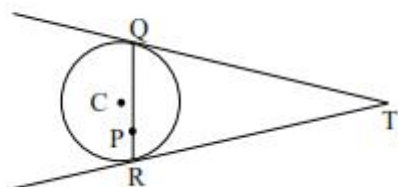
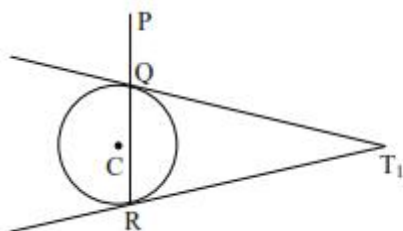
- (1) Power remains constant for the circle.
- (2) $PA \cdot PB = PC \cdot PD = PT^2$.
- **Chord of Contact:** If P is a point outside the circle and the tangents from P touch the circle in A and B, then the chord joining A and B is called as chord of contact of P with respect to the circle. The chord of contact of (x_1, y_1) with respect to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $\Rightarrow x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c = 0$.



- (1) Length of chord of contact $= 2\sqrt{r^2 - p^2}$ (p is the length of perpendicular from centre to the chord).
 - (2) Area of triangle APQ $= \frac{a\sqrt{x_1^2 + y_1^2 - a^2}}{x_1^2 + y_1^2}$.
 - (3) Equation of the chord of the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ bisected at a given point (x_1, y_1) is $x_1x + y_1y + g(x_1 + x) + f(y_1 + y) + c = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$.
- **Common chord of two circles:** The chord joining the point of intersection of two given circles is called as their common chord.



- (1) Equation of common chord to the circles $S_1 = x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$ and $S_2 = x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$ is $S_1 - S_2 = 2x(g_1 - g_2) + 2y(f_1 - f_2) + c_1 - c_2 = 0$.
 - (2) Length of common chord is $PQ = 2PM = 2\sqrt{C_1P^2 - C_2M^2}$.
- **Diameter of a circle:** The locus of the middle point of the system of parallel chord of a circle is called a diameter of the circle.
 - (1) The equation of diameter bisecting parallel chords $y = mx + c$ of the circle $x^2 + y^2 = a^2$ is $x + my = 0$.
 - **Pole and Polar:** If P is a point other than the centre in the plane of a circle and a secant line through P cuts the circle in A and B. If the tangents at A and B intersect at Q, then the locus of Q is a straight line called as the Polar of P with respect to the circle. To this polar, P is called as Pole with respect to the circle.



- (1) The equation of polar of the point (x_1, y_1) with respect to the circle $x^2 + y^2 = a^2$ is $\Rightarrow xx_1 + yy_1 = a^2$.
- (2) The equation of polar of the point (x_1, y_1) with respect to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $\Rightarrow xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$.

Some Important points on Pole and Polar:

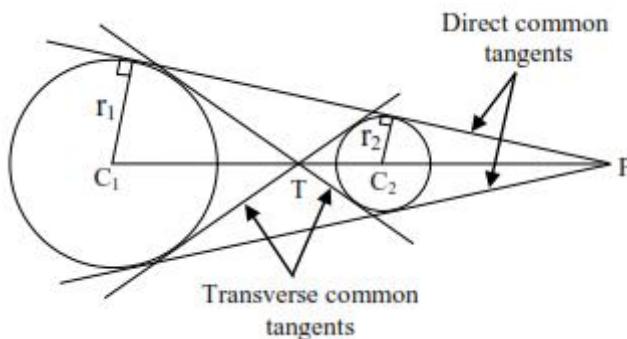
- (a) If P lies outside the circle, then polar is chord of contact of P.
 (b) If P lies on the circle, then polar is tangent at P to the circle.
 (c) If P lies inside the circle then polar completely lies outside the circle.
 (d) The polar of centre of circle with respect to the same circle does not exist.
 (e) For any diameter pole does not exist.
 (f) Polar is a straight line and it is perpendicular to the line joining centre and the point.
 (g) The polar of a point if exists is unique.
 (h) Polars of collinear points are concurrent.
 (i) Poles of concurrent line are collinear.
- (j) Pole of the line $lx + my + n = 0$ with respect to the circle $x^2 + y^2 = r^2$ is $\left(\frac{-lr^2}{n}, \frac{-mr^2}{n} \right)$.
- (k) Pole of the line $lx + my + n = 0$ with respect to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $\left(-g - \frac{lr^2}{l(-g) + m(-f) + n}, -f - \frac{mr^2}{l(-g) + m(-f) + n} \right)$.

- **Conjugate Points:** If polar of a point P with respect to a circle passes through another point Q, then the polar of Q passes through P. Two such points are called a pair of conjugate points. $P(x_1, y_1)$ and $Q(x_2, y_2)$ are conjugate points with respect to the circle $x^2 + y^2 + 2gx + 2fy + c = 0 \Leftrightarrow x_1x_2 + y_1y_2 + g(x_1 + x_2) + f(y_1 + y_2) + c = 0$.
For a point there exists an infinite number of conjugate points.
- **Conjugate Lines:** If pole of the line l_1 with respect to a circle lies on the line l_2 , then the pole of l_2 with respect to a circle lies on the line l_1 . Two such lines l_1 and l_2 are called as conjugate lines.
- Condition that the line $l_1x + m_1y + n_1 = 0$ and $l_2x + m_2y + n_2 = 0$ to be conjugate with respect to a circle $x^2 + y^2 = r^2$ is $r^2(l_1l_2 + m_1m_2) = n_1n_2$.
- Condition that the line $l_1x + m_1y + n_1 = 0$ and $l_2x + m_2y + n_2 = 0$ to be conjugate with respect to a circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is:
 $(g^2 + f^2 - c)(l_1l_2 + m_1m_2) = (l_1g + m_1f - n_1)(l_2g + m_2f - n_2)$.
 For a line there exists infinite number of conjugate lines.
- **Common Tangents to Two Circles:**
 Let two circles $(x - x_1)^2 + (y - y_1)^2 = r_1^2$ and $(x - x_2)^2 + (y - y_2)^2 = r_2^2$ with centres $C_1(x_1, y_1)$ and $C_2(x_2, y_2)$ and radii r_1 and r_2 . Then different cases may arise:

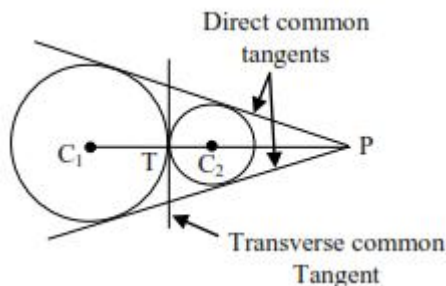
Case-1: When $C_1C_2 > r_1 + r_2$, i.e. distance between their centres greater than the sum of radii. Then four common tangents can be drawn to the two circles. Two are direct and two are transverse. The points P and T of intersection of direct common tangents and transverse common tangents always lie on the line joining the centres of two circles and divide it externally and internally in the ratio of their radii.

$$\frac{C_1P}{C_2P} = \frac{r_1}{r_2} \text{ (externally) and } \frac{C_1T}{C_2T} = \frac{r_1}{r_2} \text{ (internally).}$$

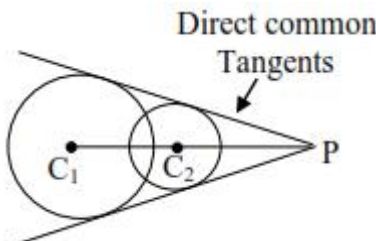
$$\text{Co-ordinate of } P = \left(\frac{r_1x_2 - r_2x_1}{r_1 - r_2}, \frac{r_1y_2 - r_2y_1}{r_1 - r_2} \right) \text{ and } T = \left(\frac{r_1x_2 + r_2x_1}{r_1 + r_2}, \frac{r_1y_2 + r_2y_1}{r_1 + r_2} \right)$$



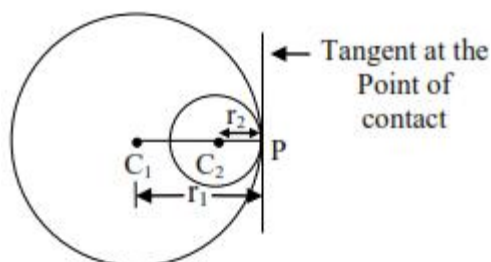
Case-2: When $C_1C_2 = r_1 + r_2$, i.e. distance between their centres equal to the sum of radii. Then two direct common tangents are real and distinct and two transverse tangents are coincident.



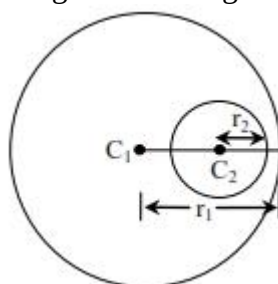
Case-3: When $C_1C_2 < r_1 + r_2$, i.e. distance between their centres less than the sum of radii. Then two direct common tangents are real and distinct and transverse tangents are imaginary.



Case-4: When $C_1C_2 = |r_1 - r_2|$, i.e. distance between their centres equal to the difference of radii. Then two direct common tangents are real and coincident and two transverse tangents are imaginary.



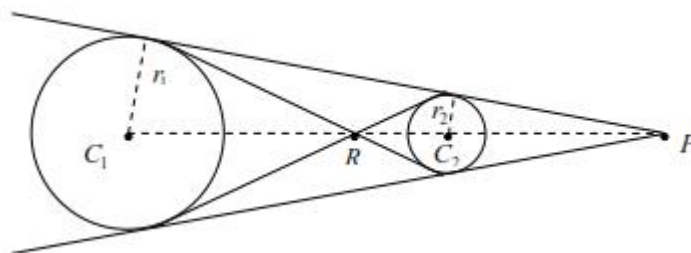
Case-5: When $C_1C_2 < |r_1 - r_2|$, i.e. distance between their centres less than the difference of radii. Then all the four tangents are imaginary.



• **Rule to find Direct Common Tangent:**

- (1) Find the co-ordinate of the centres C_1 and C_2 and radii r_1 and r_2 of two given circles.
- (2) Find the co-ordinate of the point $P(h, k)$ dividing C_1C_2 externally in the ratio $r_1 : r_2$.
- (3) Write equation of line through $P(h, k)$ as $y - k = m(x - h)$ -----→(1)
- (4) Find the two values of m , using the fact that the length of the perpendicular on the line (1) from the centre C_1 of one circle equal to its radii r_1 .
- (5) Substituting these values of m in (1), the equation of the two direct common tangents can be obtained.
- (6) The direct common tangents to two circles meet on the line joining centres C_1 and C_2 and divide it externally in the ratio of their radii $r_1 : r_2$.

• **Rule to find Transverse Common Tangents:**

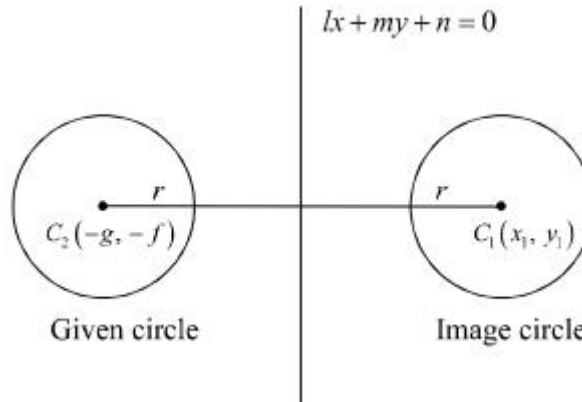


- (1) Find the co-ordinate of the centres C_1 and C_2 and radii r_1 and r_2 of two given circles.
- (2) Find the co-ordinate of the point $P(h, k)$ dividing C_1C_2 internally in the ratio $r_1 : r_2$.
- (3) Write equation of line through $P(h, k)$ as $y - k = m(x - h)$ -----→(1)
- (4) Find the two values of m , using the fact that the length of the perpendicular on the line (1) from the centre C_1 of one circle equal to its radii r_1 .

- (5) Substituting these values of m in (1), the equation of the two direct common tangents can be obtained.
- (6) The transverse common tangents also meet on the line of centres C_1 and C_2 and divide it internally in the ratio of their radii $r_1 : r_2$.

• **Image of a circle with respect to a line:**

Let circle is $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ and the line $L \equiv lx + my + n = 0$



The radius of the image circle is same as the given circle.

Let centre of image of circle is (x_1, y_1) .

Slope of $C_1C_2 \times$ slope of line $L = -1$.-----→(1)

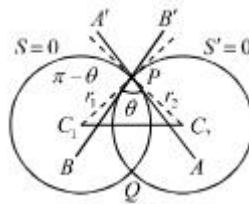
Mid point of C_1C_2 lies on line $L \equiv lx + my + n = 0$.

$$\Rightarrow l\left(\frac{x_1 - g}{2}\right) + m\left(\frac{y_1 - f}{2}\right) + n = 0. \text{-----} \rightarrow (2)$$

Solving (1) and (2) we get equation of image circle is

$$(x - x_1)^2 + (y - y_1)^2 = \left(\sqrt{g^2 + f^2 - c}\right)^2.$$

• **Angle of Intersection of two circles:**



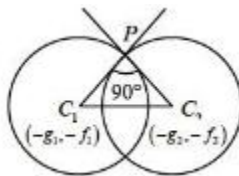
The angle of intersection between two circles $S_1 \equiv x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$ and $S_2 \equiv x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$ is defined as the angle between their tangents at their point of intersection.

Let r_1, r_2 are two radii and d be the distance between their centres then the angle of

intersection between them is given by: $\cos^{-1} = \left| \frac{r_1^2 + r_2^2 - d^2}{2r_1r_2} \right|$

$$\Rightarrow \cos^{-1} = \left| \frac{2(g_1g_2 + f_1f_2) - (c_1 + c_2)}{2\sqrt{g_1^2 + f_1^2 - c_1}\sqrt{g_2^2 + f_2^2 - c_2}} \right|.$$

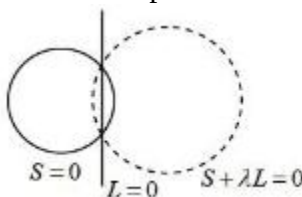
• **Condition of orthogonality:**



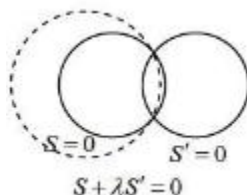
If the angle of intersection of the two circle is a right angle i.e. $\angle = 90^\circ$, then such circles are called orthogonal and the condition of two circles $S_1 \equiv x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$ and $S_2 \equiv x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$ are orthogonal to each other if $2(g_1g_2 + f_1f_2) - (c_1 + c_2) = 0 \Rightarrow 2(g_1g_2 + f_1f_2) = (c_1 + c_2)$.

• **Family of Circles:**

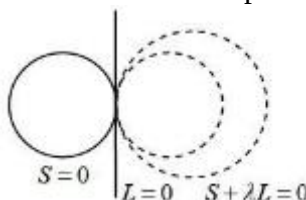
- (1) The equation of circles passing through the point of intersection of circle $S = 0$ and line $L = 0$ is $S + \lambda L = 0$ where λ is a parameter.



- (2) The equation of family of circles passing through the point of intersection of two given circles $S_1 = 0$ and $S_2 = 0$ is given as $S_1 + \lambda S_2 = 0$ where $\lambda \neq -1$ is a parameter. It is better to find $S_1 + \lambda(S_1 - S_2) = 0$.



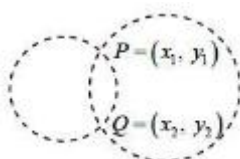
- (3) The equation of family of circles touching the circle $S = 0$ and the line $L = 0$ at their point of contact P is $S + \lambda L = 0$ where λ is a parameter.



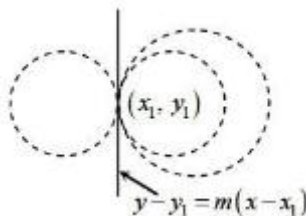
- (4) The equation of family of circles passing through two given points $P(x_1, y_1)$ and

$$Q(x_2, y_2) \text{ can be written as } (x - x_1)(x - x_2) + (y - y_1)(y - y_2) + {}^a \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0,$$

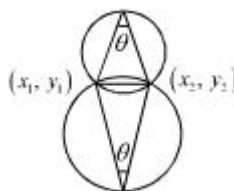
where a is a parameter.



- (5) The equation of family of circles, which touch $(y - y_1) = m(x - x_1)$ at (x_1, y_1) for any finite m is $(x - x_1)^2 + (y - y_1)^2 + {}^a \{(y - y_1) - (x - x_1)\} = 0$ and if m is infinite then family of circles is $(x - x_1)^2 + (y - y_1)^2 + {}^a (x - x_1) = 0$, where a is a parameter.



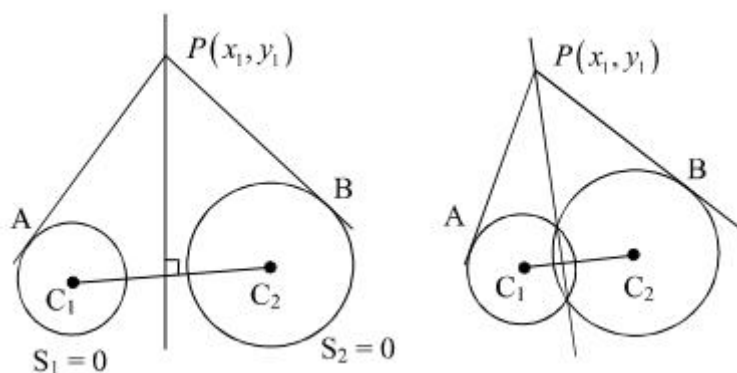
- (6) For the given diagram:



Equation of circles is:

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) \pm \cot^{-1} \{(x - x_1)(y - y_2) - (x - x_2)(y - y_1)\} = 0$$

• **Radical Axis:**



The radical axis of two circles is the locus of a point which moves such that the lengths of the tangents drawn from it to the two circles are equal.

The equation of the radical axis of the two circles is $S_1 - S_2 = 0$ i.e.

If $S_1 \equiv x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$ and $S_2 \equiv x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$ are two equation of circles then equation of radical axis is:

$$S_1 - S_2 = 0$$

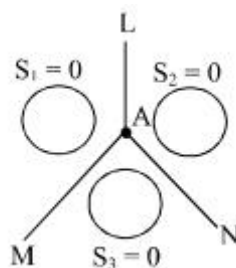
$$\Rightarrow x^2 + y^2 + 2g_1x + 2f_1y + c_1 - x^2 - y^2 - 2g_2x - 2f_2y - c_2 = 0$$

$$\Rightarrow 2x(g_1 - g_2) + 2y(f_1 - f_2) + c_1 - c_2 = 0, \text{ which represents a straight line.}$$

• **Properties of Radical Axis:**

- (1) The radical axis and common chord are identical for two intersecting circles.
- (2) The radical axis of two circles is perpendicular to the line joining their centres.
- (3) Radical Centre: The radical axis of three circles taken in pairs meet at a point, called the radical centre of the circles. Co-ordinates of radical centre can be found by solving the equations: $S_1 = S_2 = S_3 = 0$.
- (4) The radical centre of three circles described on the sides of a triangle as diameters is the orthocenter of the triangle.
- (5) If two circles cut a third circle orthogonally, then the radical axis of the two circles pass through the centre of the third circle.

• **Radical Centre:**



The radical axes of three circles, taken in pairs meet at a point, which is called as their radical centre.

Let the three circles are:

$$S_1 = 0 \text{ -----} \rightarrow (1)$$

$$S_2 = 0 \text{ -----} \rightarrow (2)$$

$$S_3 = 0 \text{ -----} \rightarrow (3)$$

Let the straight lines AL and AM meet in A.

Equation of any straight line passing through A is $(S_1 - S_2) + \lambda(S_3 - S_1) = 0$, where λ is a parameter.

For $\lambda = 1$, this equation is $(S_3 - S_2) = 0$, which is equation of AN.

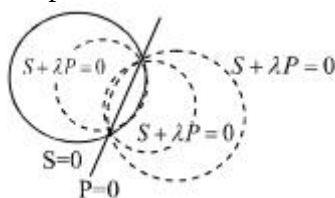
Thus the third radical axis also passes through the point where the straight lines AL and AM meet. In the figure A is the radical centre.

• **Properties of Radical Centre:** (co-axial system of circles)

- (1) The equation of system of co-axial circles, when the equation of the radical axis and of one circle of the system are:

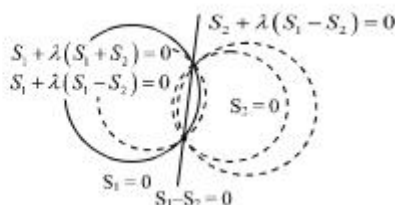
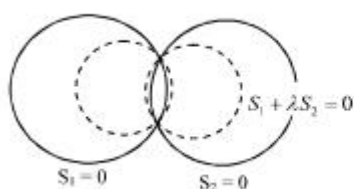
Let circle is $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ and the line $P \equiv lx + my + n = 0$ is

$S + \lambda P = 0$, where λ is a parameter.



- (2) The equation of co-axial system of circles, where the equation of any two circles of the system are:

$S_1 \equiv x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$ and $S_2 \equiv x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$ is $S_1 + \lambda S_2 = 0$, where $\lambda \neq -1$ is a parameter.



- (3) The equation of system of co-axial circles in the simplest form is:
 $x^2 + y^2 + 2gx + c = 0$, where g is a variable and c is a constant.

- **Limiting Point:** Limiting point of a system of co-axial circles are the centres of the point circles belonging to the family (circles whose radii are zeroes are called point circles).

Let the circle is $x^2 + y^2 + 2gx + c = 0 \rightarrow (1)$, where g is a variable and c is a constant.

Centre of (1) is $(-g, 0)$.

Radius of (1) is $\sqrt{g^2 - c}$.

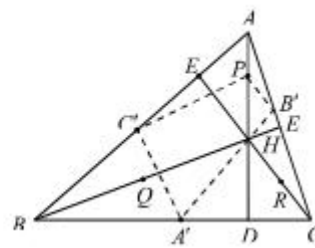
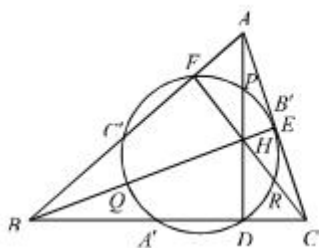
Let $\sqrt{g^2 - c} = 0 \Rightarrow g = \pm\sqrt{c}$.

Thus we get the two limiting points of the given co-axial system as $(\sqrt{c}, 0)$ and $(-\sqrt{c}, 0)$.

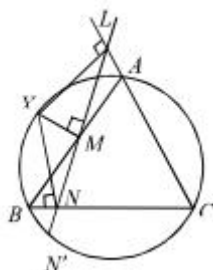
Clearly the limiting points are real and distinct, real and coincident or imaginary according as $c > 0$, $c = 0$, $c < 0$.

- **Nine Point Circles:** The circles through the mid points of the sides of a triangle, passes through the feet of the altitudes and the mid points of the line joining orthocenter to the vertices. The circle is called as nine-point circle of the triangle.

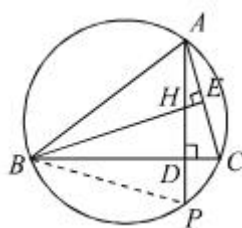
- (1) The radius of the nine point circle is half the radius of the circumcircle of the triangle ABC.
- (2) Centre is midpoint of the line segment joining orthocenter and circumcentre.



- **Simpson's Line:** The feet L, M, N of the perpendicular on the sides BC, CA, AB of any triangle ABC from any point X on the circumcircle of the triangle are collinear. The line LMN is called as Simpson's line or Pedal line of the point X with respect to triangle ABC.



If H is the orthocenter of triangle ABC and AH produced meets BC at D and the circumcircle of triangle at P, then $HD = DP$.



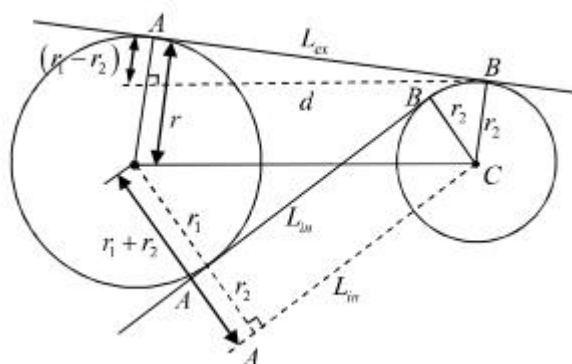
- **Inverse point:** C is centre and r is the radius of a circle. If P is a point, then the point of intersection of the lines CP and polar of P is called as inverse point of P with respect to the circle.

Some Important points on Inverse Points:

- If Q is inverse point of P, then P is the inverse point of Q and P, Q is a pair of inverse points.
- If P lies inside a circle, then its inverse point lies outside the circle.
- If P lies on the circle then inverse point of P is itself.
- Inverse point of centre does not exist.
- Every pair of inverse points is a pair of conjugate points, but the converse is not true.
- Inverse point of a point if exists is unique.
- If C is centre, r is the radius of a circle and P, Q are pair of inverse points then $CP.PQ = r^2$.
- Pair of inverse points P and Q, lies on the same side of the centre.
- The inverse point of $P(x_1, y_1)$ with respect to circle $x^2 + y^2 = r^2$ is $\left(\frac{r^2 x_1}{x_1^2 + y_1^2}, \frac{r^2 y_1}{x_1^2 + y_1^2} \right)$.
- The inverse point of $P(0,0)$ with respect to circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $\left(\frac{-gc}{g^2 + f^2}, \frac{-fc}{g^2 + f^2} \right)$.

- Equation of chord of circle $x^2 + y^2 + 2gx + 2fy + c = 0$ whose mid point is (x_1, y_1) is $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$. And its length is $2\sqrt{x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c}$.
- Equation of the pair of tangent from (x_1, y_1) to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $(xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c)^2 = (x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c)(x^2 + y^2 + 2gx + 2fy + c)$
- Equation of the pair of tangent from $(0, 0)$ to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $(gx + fy)^2 = c(x^2 + y^2 + 2gx + 2fy + c)$.
- Locus of point of intersection of perpendicular tangents to the circle:
 - $x^2 + y^2 = r^2$ is $x^2 + y^2 = 2r^2$.
 - $(x - x_1)^2 + (y - y_1)^2 = r^2$ is $(x - x_1)^2 + (y - y_1)^2 = 2r^2$.
 - $x^2 + y^2 + 2gx + 2fy + c = 0$ is $x^2 + y^2 + 2gx + 2fy + c = g^2 + f^2 - c$.
- Locus of point of intersection of perpendicular tangents one to each of the circle:
 - $x^2 + y^2 = a^2$, $x^2 + y^2 = a^2$ is $x^2 + y^2 = a^2 + b^2$.
 - $(x - x_1)^2 + (y - y_1)^2 = a^2$, $(x - x_1)^2 + (y - y_1)^2 = a^2$ is $(x - x_1)^2 + (y - y_1)^2 = a^2 + b^2$.
 - $x^2 + y^2 + 2gx + 2fy + c_1 = 0$, $x^2 + y^2 + 2gx + 2fy + c_2 = 0$ is $x^2 + y^2 + 2gx + 2fy + c_1 = g^2 + f^2 - c_2$.
- Locus of point of intersection of two tangents which includes an angle θ , to the circle $x^2 + y^2 = r^2$ is $x^2 + y^2 = r^2 \operatorname{cosec}^2(\theta/2)$.
- Locus of mid points of chords of the $x^2 + y^2 = r^2$ which subtend angle θ at centre is $x^2 + y^2 = r^2 \cos^2(\theta/2)$.
- Locus of mid points of chords of the $x^2 + y^2 = r^2$ of length $2k$ units is $x^2 + y^2 = r^2 - k^2$.
- The locus of the mid points of the chords of the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ passing through a given point is a circle with the given point and the centre of the given circle are the end points of the diameter.
- Locus of mid points of chords of the $x^2 + y^2 = r^2$ which are at a distance of k units from the centre is Locus of mid points of chords of the $x^2 + y^2 = k^2$.
- If the circle $S = 0$, and $L = 0$ is a line intersects the circle in A and B, then equation of a circle that passes through A and B is $S + \lambda L = 0$.
- Area of a quadrilateral formed by the two tangents drawn from an external points (x_1, y_1) to a circle $x^2 + y^2 + 2gx + 2fy + c = 0$ and a pair of radii through their point of contact is $r\sqrt{x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c}$.
- The length of chord of contact of the point (x_1, y_1) with respect to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $2r\sqrt{\frac{x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c}{x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c + g^2 + f^2 - c}}$.
- If C_1 and C_2 are centre of two circles, whose radii are r_1 and r_2 where $r_1 > r_2$ then number of common tangents of these circle:

- (a) ZERO if $r_1 - r_2 > C_1C_2$.
- (b) ONE if $r_1 - r_2 = C_1C_2$.
- (c) TWO if $r_1 - r_2 < C_1C_2 < r_1 + r_2$.
- (d) THREE if $C_1C_2 = r_1 + r_2$.
- (e) FOUR if $C_1C_2 > r_1 + r_2$.
- If C_1 and C_2 are centre of two circles, whose radii are r_1 and r_2 where $r_1 > r_2$, then they will touch each other if $C_1C_2 = r_1 \pm r_2$.
 - External Centre of Similitude:** The point of intersection of direct common tangents of two circles is called as external centre of similitude. If C_1 and C_2 are centres and r_1 and r_2 are radii, then external centre of similitude divides the line joining C_1 and C_2 in the ratio $r_1 : r_2$ externally.
 - Internal Centre of Similitude:** The point of intersection of transverse common tangents of two circles is called as internal centre of similitude. If C_1 and C_2 are centres and r_1 and r_2 are radii, then internal centre of similitude divides the line joining C_1 and C_2 in the ratio $r_1 : r_2$ internally.
 - Lengths of Common Tangents:**



- If d is the distance between centres of two circles with radii r_1 and r_2 , then length of direct common tangent of two circles is $\sqrt{d^2 - (r_1 - r_2)^2}$.
- If d is the distance between centres of two circles with radii r_1 and r_2 , then length of transverse common tangent of two circles is $\sqrt{d^2 - (r_1 + r_2)^2}$.
